(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

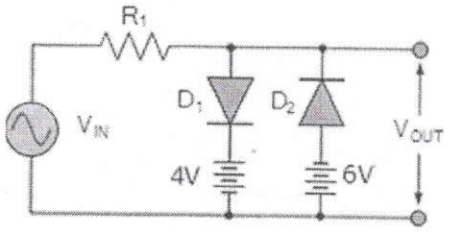
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

___Third_____Semester, November, 2022

Paper Code: __ECC-215_____ **Subject: ___Electronics-I_____**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Explain effect of temperature on intrinsic and extrinsic semiconductors.	2	1
1 (b)	Explain zener and avalanche breakdown mechanism.	3	1
1 (c)	What are I_{CBO} and I_{CEO} currents in BJT Transistor configurations.	2	2
1 (d)	Write differences between FET and BJT Transistors.	3	2
2 (a)	Explain the working of LED and Photodiode.	5	1
2 (b)	In a transistor connected in the common base configuration, α=0.95, I_E =2 mA. Calculate the values of I_C and I_B	5	2
3 (a)	Find the resistivity of an intrinsic semiconductor with intrinsic concentration of 1.5×10^{19} per m^3. The mobilities of electrons and holes are 0.20 m^2/ V-s and 0.10 m^2/ V-s.	5	1
3 (b)	Explain in detail input and output characteristics of common base transistor configuration.	5	2
4(a)	Draw the output waveform for the circuit given below  Also draw diagram for positive clamper circuit.	5	1
4 (b)	Draw hybrid model for CE configuration of NPN BJT transistor.	5	2

*Note- Necessary data may be assumed wherever necessary and same may be clearly indicated.